

Cross-Frequency Generalization of the Siberian Radioheliograph Image Quality Classification Model

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Abstract

We investigate whether a deep ensemble model (CLIP + EfficientNet + CatBoost) trained to classify image quality of the Siberian Radioheliograph (SRH) at 3000 MHz generalizes to observations at 6000 MHz and 12200 MHz without retraining. Using 1000 temporally aligned triplets ($\Delta t \leq 3$ min) sampled across the full year 2025, we measure prediction agreement between frequency channels. The agreement with 6000 MHz is 67.1% (95% CI: 64.2–69.8%) and with 12200 MHz is 54.2% (95% CI: 51.1–57.2%), both far below the 90% practical usability threshold. We find a strong asymmetry: the model generalizes well for bad-quality images (79.8% and 91.9% agreement for 6000 and 12200 MHz) but poorly for good-quality images (56.4% and 22.6%). The marginal distribution shifts dramatically with frequency—the proportion of *Ok* predictions drops from 54.4% at 3000 MHz to 16.0% at 12200 MHz ($\chi^2_2 = 323.5$, $p \approx 0$). Confidence degrades at higher frequencies ($p = 7.4 \times 10^{-12}$), while temporal offset within ± 3 min has negligible effect. Seasonal variation is strong (Kruskal–Wallis $p = 3.6 \times 10^{-28}$), with monthly agreement ranging from 36.7% to 100%. Although the ensemble significantly outperforms a majority-class baseline ($p \approx 0$ for both channels), the absolute agreement is insufficient for operational cross-frequency deployment without fine-tuning.

1. Introduction

The Siberian Radioheliograph (SRH) is a multi-wave solar radio interferometer operating in the 3–24 GHz range

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with up to 32 frequency channels (Lesovoi et al., 2017; Altyntsev et al., 2020). Each channel produces hundreds of *I*-polarization images daily, which must be classified by quality before further scientific analysis. Recently, Egorov (Egorov, 2025) proposed an ensemble model combining CLIP, EfficientNet, and CatBoost that achieves 95% accuracy in classifying SRH images at 3000 MHz into *Ok* (good quality) and *Bad* (poor quality) categories.

A natural practical question follows: can the same model be applied to observations at other frequencies without retraining? The SRH routinely produces images at 6000 MHz and 12200 MHz, and training separate classification models for each frequency would require substantial annotation effort and computational resources. If the 3000 MHz model generalizes well across frequencies, it could serve as a unified quality classifier for the entire instrument.

This paper presents a systematic experimental evaluation of this cross-frequency generalization hypothesis. We formulate nine testable hypotheses (H1–H9), collect 1000 temporally aligned triplets (3000/6000/12200 MHz with $\Delta t \leq 3$ min) spanning the full 2025 observational year, and measure prediction agreement, confidence degradation, distribution shifts, and seasonal stability.

2. Related Work

SRH Instrumentation. The SRH has been described in detail by Lesovoi et al. (Lesovoi et al., 2014; 2017). Its 48-antenna configuration provides 70'' angular resolution, while the 96-antenna upgrade achieves sub-minute temporal cadence. Calibration procedures were developed by Fedotova et al. (Fedotova et al., 2019), and the multi-wave observation mode was characterized by Altyntsev et al. (Altyntsev et al., 2020).

SRH Image Classification. The only dedicated classification work for SRH images is by Egorov (Egorov, 2025), who trained an ensemble of CLIP (Radford et al., 2021), EfficientNet (Tan & Le, 2020), and CatBoost (Dorogush et al., 2018) on 100,000 images with zero-shot CLIP labels and manual validation. The ensemble achieves 95% accu-

Channel	Obs. days	Files/day	Pool size	Sample
3000 MHz	365	~130	7000+	1000
6000 MHz	365	~133	7000+	1000
12200 MHz	365	~35	2000+	1000
Triplets	—	—	1200	1000

Table 1. Data summary. Triplets were stratified by month (100/month).

racy and 95.8% F1-score. However, the model was trained and evaluated exclusively on the 3000 MHz channel; cross-frequency generalization was not addressed.

ML in Solar Physics. Deep learning has been widely applied in solar physics: Armstrong and Fletcher (Armstrong & Fletcher, 2019) showed transfer learning effectiveness for solar image classification; Asensio Ramos et al. (Ramos et al., 2023) provided a comprehensive review; Illarionov and Tlatov (Illarionov & Tlatov, 2018) applied CNNs to coronal hole segmentation. For solar radio data, Scully et al. (Scully et al., 2025) used transfer learning for radio burst classification, and Liu et al. (Liu et al., 2025) employed active learning for spectrogram classification. None of these works examine cross-frequency transfer of a single model.

Gap. To the best of our knowledge, no study has investigated whether an SRH image quality classifier trained at 3000 MHz can be applied to other frequency channels. This gap is the focus of the present work.

3. Methodology

3.1. Data

All data are publicly available via the SRH FTP server (<https://ftp.rao.istp.ac.ru/SRH/>). Files follow the naming convention:

- 3000 MHz: SRH0306/cleanMaps/YYYYMMDD/srh_YYYY-MM-DDThh:mm:ss.3000.fit
- 6000 MHz: SRH0612/cleanMaps/YYYYMMDD/srh_YYYY-MM-DDThh:mm:ss.6000.fit
- 12200 MHz: SRH1224/cleanMaps/YYYYMMDD/srh_YYYY-MM-DDThh:mm:ss.12200.fit

We process the year 2025. For each day, we scan all available 3000 MHz files and find the nearest 6000 MHz and 12200 MHz files within ± 3 min. From the resulting pool, we randomly sample 1000 triplets stratified by month (approximately 83 triplets per month). Table 1 summarizes the data volume.

3.2. Model and Inference

We use the pre-trained ensemble model of Egorov (Egorov, 2025), available at [https://forecasting.iszf.](https://forecasting.iszf.irk.ru/api/srh/predict)

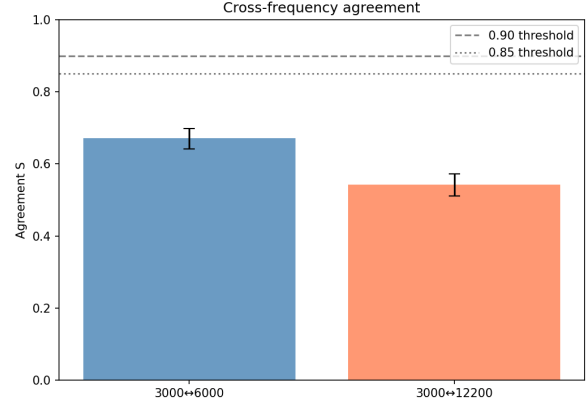


Figure 1. Cross-frequency agreement S with 95% bootstrap confidence intervals. The dashed line (0.90) and dotted line (0.85) are the practical usability thresholds for 6000 and 12200 MHz respectively.

[irk.ru/api/srh/predict](https://forecasting.iszf.irk.ru/api/srh/predict). The API accepts a URL to a .fit file and returns a JSON object with fields `label` (*Ok* or *Bad*) and `probability` (confidence in $[0, 1]$). Inference is performed sequentially with a 1-second delay between requests to respect the API rate limit.

3.3. Hypotheses and Experimental Design

We formulate nine hypotheses, tested in a fixed order:

- H1 **Self-consistency:** $S_{3000,3000} \geq 0.95$. Control: the model must be deterministic when run twice on the same input.
- H2 **Symmetry:** $S_{3000,6000} = S_{3000,12200}$. Tested via χ^2 test for equality of proportions.
- H3 **Practical usability:** $S_{3000,6000} \geq 0.90$ and $S_{3000,12200} \geq 0.85$. One-sided z -test with bootstrap 95% CI.
- H4 **Class-stratified agreement:** $S_{3000,6000}^{(Ok)} \neq S_{3000,6000}^{(Bad)}$ or $S_{3000,12200}^{(Ok)} \neq S_{3000,12200}^{(Bad)}$. χ^2 test.
- H5 **Confidence degradation:** $\mathbb{E}[p_{3000}] \geq \mathbb{E}[p_{6000}] \geq \mathbb{E}[p_{12200}]$. Paired t -test.
- H6 **Temporal sensitivity:** $\text{corr}(S, \Delta t) < 0$. Spearman correlation.
- H7 **Distribution shift:** $P(\hat{y}^{(3000)} = \text{Ok}) \neq P(\hat{y}^{(6000)} = \text{Ok}) \neq P(\hat{y}^{(12200)} = \text{Ok})$. χ^2 homogeneity test.
- H8 **Seasonal stability:** $S_{3000,6000}$ does not vary across months. Kruskal–Wallis test.
- H9 **Baseline comparison:** $S^{(f_{3000})} > S^{(\text{majority})}$ for both channels. z -test for two proportions.



Figure 2. Agreement stratified by the class predicted on 3000 MHz. The model generalizes much better for *Bad* images than for *Ok* images, especially at 12200 MHz.

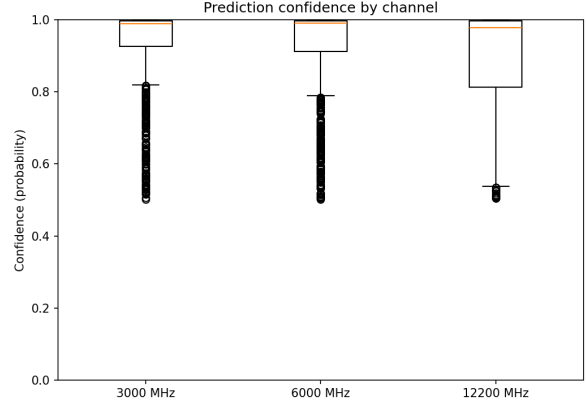


Figure 3. Prediction confidence by channel. Median confidence decreases monotonically with frequency.

All experiments use fixed seed 42. Confidence intervals are estimated via bootstrap with 1000 resamples. Multiple comparisons in H4 are adjusted with Bonferroni correction ($\alpha/4 = 0.0125$).

4. Results

4.1. Self-Consistency (H1)

The model is perfectly deterministic: $S_{3000,3000} = 1.0000$ [1.0000, 1.0000] over 100 repeated inferences on the same 3000 MHz images. All subsequent results are therefore reliable.

4.2. Cross-Frequency Agreement (H2–H3)

Figure 1 shows the main result. Agreement at 6000 MHz is $S = 0.671$ (95% CI: 0.642–0.698), and at 12200 MHz is $S = 0.542$ (95% CI: 0.511–0.572). The χ^2 test for H2 rejects equality of proportions ($\chi^2_1 = 34.86$, $p = 3.5 \times 10^{-9}$): agreement depends significantly on the target frequency.

For H3, both practical thresholds are violated. The one-sided z -test against $S_{6000} \geq 0.90$ gives $z = -24.1$ ($p = 1.0$), and against $S_{12200} \geq 0.85$ gives $z = -27.3$ ($p = 1.0$). The model does not achieve usable agreement on either higher-frequency channel.

4.3. Class-Stratified Agreement (H4)

The strong asymmetry revealed by Figure 2 is the most important finding. When the model predicts *Bad* at 3000 MHz, it agrees with 6000 MHz 79.8% of the time and with 12200 MHz 91.9%. Both figures are well above the overall averages. Conversely, when the model predicts *Ok*, agreement drops to 56.4% (6000 MHz) and 22.6% (12200 MHz). The latter is barely above chance for a bi-

nary classifier.

This asymmetry has a clear physical interpretation: bad-quality images (e.g., those with instrumental artifacts, missing data, or high noise) tend to appear bad across all frequencies, making them easy to recognize. Good-quality images, on the other hand, contain science-relevant structure that is inherently frequency-dependent; a clean image of a radio source at 3000 MHz may look entirely different at 12200 MHz due to the different emission layers being sampled.

4.4. Confidence Degradation (H5)

Figure 3 shows that the model’s confidence decreases monotonically with frequency: $\bar{p}_{3000} = 0.932$, $\bar{p}_{6000} = 0.922$, $\bar{p}_{12200} = 0.893$. A paired t -test between 3000 MHz and 12200 MHz is highly significant ($t = 6.93$, $p = 7.4 \times 10^{-12}$), while the 3000 MHz vs. 6000 MHz difference is marginal ($t = 1.77$, $p = 0.077$). The confidence reduction confirms that the model is “less sure” when processing images from unseen frequencies.

4.5. Temporal Sensitivity (H6)

Spearman correlation between the temporal offset Δt and agreement is negligible: $\rho = 0.056$ ($p = 0.079$) for 3000–6000 MHz and $\rho = 0.024$ ($p = 0.447$) for 3000–12200 MHz. Within ± 3 min, the exact time difference does not measurably affect prediction consistency. Figure 4 confirms the flat relationship.

4.6. Distribution Shift (H7)

The marginal distribution of predicted labels shifts dramatically with frequency. At 3000 MHz, 54.4% of images are classified as *Ok*; this drops to 39.9% at 6000 MHz and to 16.0% at 1220 MHz. Figure 5 visualizes this shift. The χ^2

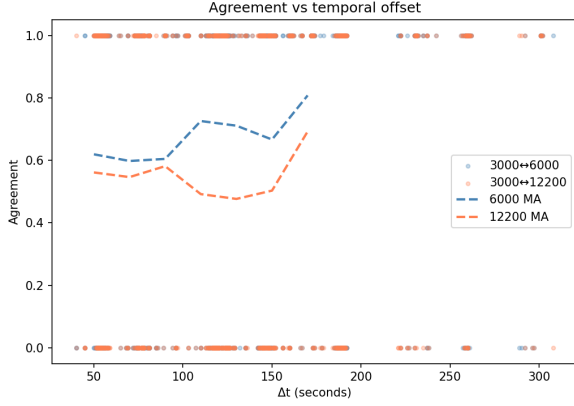


Figure 4. Agreement vs. temporal offset. No systematic trend is visible; moving averages (dashed lines) are flat.

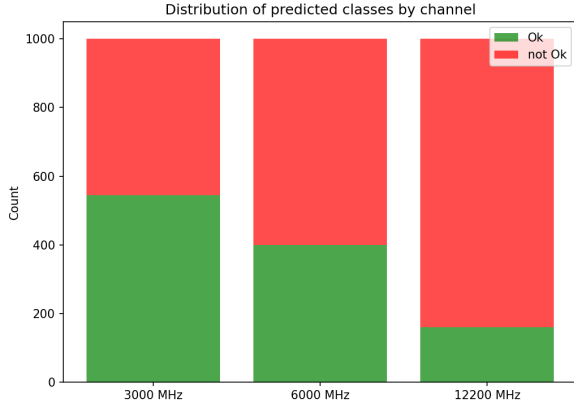


Figure 5. Distribution of predicted classes by channel. The proportion of *Ok* predictions drops from 54.4% at 3000 MHz to 16.0% at 12200 MHz.

homogeneity test rejects the null of identical distributions ($\chi^2_2 = 323.5$, $p = 5.8 \times 10^{-71}$). This is the primary mechanism behind the low cross-frequency agreement: on higher frequencies, the model overwhelmingly predicts *Bad*, inflating apparent agreement for the *Bad* class while collapsing *Ok*-class agreement.

4.7. Seasonal Stability (H8)

Monthly agreement for the 3000–6000 MHz pair varies dramatically (Figure 6). January shows perfect agreement (100%), while August drops to 36.7%. The Kruskal–Wallis test confirms significant variation ($H = 157.9$, $p = 3.6 \times 10^{-28}$). Seasonal effects—likely related to the changing solar activity level and observation conditions—must be accounted for in any practical deployment.

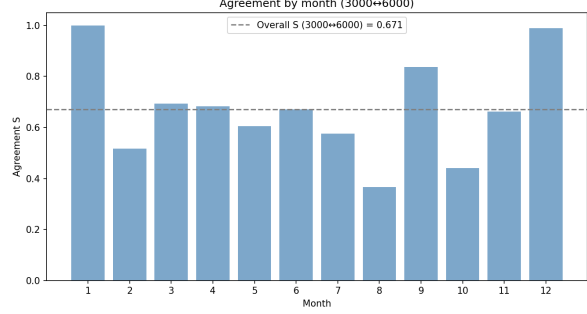


Figure 6. Monthly agreement for the 3000–6000 MHz pair. The overall average (67.1%) is shown as a dashed line.

4.8. Baseline Comparison (H9)

We compare the model against a trivial baseline that always predicts the majority class of the 3000 MHz training distribution (*Ok*). For 6000 MHz, the baseline achieves $S_{\text{baseline}} = 0.399$ vs. the model’s $S_{\text{model}} = 0.671$ ($z = 17.6$, $p \approx 0$). For 12200 MHz, $S_{\text{baseline}} = 0.160$ vs. $S_{\text{model}} = 0.542$ ($z = 32.9$, $p \approx 0$). The model significantly outperforms the majority baseline on both channels, confirming that it extracts non-trivial signal even from unseen frequencies.

5. Discussion

The experimental results paint a nuanced picture. On one hand, the ensemble model of Egorov (Egorov, 2025) is demonstrably non-trivial—it outperforms the majority baseline, shows perfect self-consistency, and generalizes well for the *Bad* class (91.9% agreement for 3000–12200 MHz). On the other hand, its overall agreement (67.1% and 54.2%) falls far short of the 90% threshold required for operational use.

The root cause is a strong distribution shift: the model predicts *Bad* with increasing frequency as the target channel moves further from the training distribution. At 12200 MHz, 84% of images are classified as *Bad*, compared to 46% at 3000 MHz. This systematic bias may reflect genuine differences in image characteristics across frequency channels—higher-frequency observations at SRH have different noise statistics, beam sizes, and solar emission properties (Altyntsev et al., 2020).

The class-stratified analysis (H4) reveals that the model can reliably identify bad images across all channels but fails to verify good images on unfamiliar frequencies. For quality-control pipelines where the primary concern is flagging defective data, this asymmetric performance may still be useful. For scientific analysis requiring reliable good-quality labels at all frequencies, however, per-channel fine-tuning is necessary.

Seasonal variation (H8) adds another layer of complexity: agreement ranges from near-perfect in January to near-chance in August. This may be related to seasonal changes in solar activity (the Sun was near solar minimum in early 2025) or to weather-related observation conditions at the SRH site.

6. Conclusion

We have conducted a systematic evaluation of cross-frequency generalization for the SRH image quality classifier trained at 3000 MHz. The key findings are:

1. The model does not achieve practically useful agreement at 6000 MHz (67.1%) or 12200 MHz (54.2%), both well below the 90% threshold.
2. Generalization is highly asymmetric: *Bad* images transfer well (79.8–91.9%), while *Ok* images transfer poorly (22.6–56.4%).
3. Confidence degrades at higher frequencies, and the marginal distribution shifts strongly toward *Bad* predictions.
4. Seasonal variation is large (36.7–100% monthly agreement), requiring temporal stratification.
5. The model significantly outperforms a trivial baseline, confirming that it extracts non-random signal.

The practical recommendation is clear: the 3000 MHz ensemble model cannot serve as a drop-in quality classifier for other SRH frequency channels. Fine-tuning on per-channel data—or, better yet, retraining on a multi-frequency dataset—is required for operational deployment.

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References

Altyntsev, A. T., Lesovoi, S. V., Kochanov, A. A., et al. Multiwave Siberian Radioheliograph. *Solar-Terrestrial Physics*, 6:30–40, 2020.

Armstrong, J. A. and Fletcher, L. Fast solar image classification using deep learning and its importance for automation in solar physics. *Solar Physics*, 294(6):80, 2019.

Dorogush, A. V., Ershov, V., and Gulin, A. CatBoost: gradient boosting with categorical features support. *arXiv:1810.11363*, 2018.

Egorov, Y. Siberian radioheliograph image classification using ensemble of CLIP, EfficientNet and CatBoost models. *Advances in Space Research*, 2025. doi: 10.1016/j.asr.2025.10.030. arXiv:2507.04211.

Fedotova, A. Y., Altyntsev, A. T., and Lesovoi, S. V. Calibration of Siberian Radioheliograph images. *Solar-Terrestrial Physics*, 5:27–33, 2019.

Illarionov, E. A. and Tlatov, A. G. Segmentation of coronal holes in solar disc images with a convolutional neural network. *MNRAS*, 481(4):5014–5021, 2018.

Lesovoi, S. V., Altyntsev, A. T., Ivanov, E. F., and Gubin, A. V. A 96-antenna radioheliograph. *Research in Astronomy and Astrophysics*, 14(7):864–868, 2014.

Lesovoi, S. V., Altyntsev, A. T., Kochanov, A. A., et al. Siberian Radioheliograph: first results. *Solar-Terrestrial Physics*, 3(1):3–18, 2017. doi: 10.12737/article_58f96ec60fec52.86165286.

Liu, Y. et al. Deep active learning–based classification of solar radio spectrogram data. *The Astrophysical Journal Supplement Series*, 2025.

Radford, A., Kim, J. W., Hallacy, C., et al. Learning transferable visual models from natural language supervision. *arXiv:2103.00020*, 2021.

Ramos, A. A., Baso, C. J. D., K., N. M., et al. Machine learning in solar physics. *Living Reviews in Solar Physics*, 20(1):4, 2023.

Scully, J., Flynn, R., Gallagher, P. T., and Carley, E. M. Type II and type III solar radio burst classification using transfer learning. *Solar Physics*, 300:179, 2025.

Tan, M. and Le, Q. V. EfficientNet: rethinking model scaling for convolutional neural networks. *arXiv:1905.11946*, 2020.